

Hybrid Model and Ensemble for Inflation Forecasting: A Machine Learning Approach

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Abstract—An inflation index is a statistical tool that measures the change in the general price level of goods and services in an economy over a specific period, typically used to gauge the inflation rate. Understanding such an index is vital because it measures the overall price level changes in an economy over time, influencing policy decisions, investments, and the real value of money, and thus, the general economic well-being of society. This article presents a study on the forecast of the official inflation index in Brazil, IPCA, using a hybrid system with bagging. The proposed model combines a linear approach (ETS) with the bagging technique applied to residual forecasting. Thirteen relevant methods mentioned in the literature were evaluated, including widely used models such as ARIMA, ETS, MLP, SVR, RF, and various hybrid combinations. The experimental results show that the proposed hybrid model (ETS+Bagging) outperformed individual models and other hybrid systems. Furthermore, the bagging approach significantly improved the results compared to the ARIMA and ETS models, as measured by metrics such as mean squared error (MSE), root means squared error (RMSE), and mean absolute error (MAE). The results indicate that the hybrid system with bagging, especially the proposed ETS+Bagging model, has the potential to enhance inflation forecasts using IPCA data.

Index Terms—Time Series, Hybrid intelligent systems, Bagging, Inflation, Forecasting,

I. INTRODUCTION

Inflation forecasts are crucial in guiding financial institutions, private sector decision-making, and other macroeconomic indicators such as unemployment and gross domestic product (GDP). This is because inflation directly impacts the real wages of individuals, eroding their purchasing power and affecting the demand for goods and services. Moreover, it influences the prices of production inputs, potentially impeding economic growth. Consequently, gaining an accurate understanding of price levels in the economy has become an utmost priority, leading to extensive research efforts in recent years [1], [2]. However, due to the dynamic nature of prices and the delayed effects of monetary policy on the economy, policymakers must take proactive measures to address inflationary or deflationary pressures [3].

The broad national consumer price index (IPCA, *Índice nacional de Preços ao Consumidor Amplo*), created in 1979 and measured by the Brazilian Institute of Geography and Statistics (IBGE), is used in Brazil as the official benchmark

indicator to measure inflation [4]. The IPCA compares the price changes of items consumed by households with incomes ranging from 1 to 40 minimum wages in different regions of the country and groups them into categories such as food and beverages, housing, household items, clothing, transportation, health and personal care, personal expenses, education, and communication [5].

In inflation forecasting, it is common to use traditional linear models such as Auto Regressive Integrated Moving Average (ARIMA), Exponential Smoothing (ETS), and Box-Jenkins time series models [6]–[8]. These models have gained popularity due to the well-established methodology by Box-Jenkins for selecting appropriate models and their ability to handle non-stationary data. However, it is essential to highlight that linear models like ARIMA and ETS may have limitations in capturing non-linear patterns in time series, as emphasized by Zhang et al. [9]. Given that real-world data often exhibit a combination of linear and non-linear patterns and that the dynamic behavior of macroeconomic series involves complex phenomena, relying solely on models like ARIMA or ETS can lead to inaccurate predictions.

Therefore, to overcome the limitations of linear forecasting techniques, machine learning models such as feed-forward multilayer perceptron (MLP) neural networks, support vector regression (SVR), and random forest (RF), have emerged as valuable tools for handling non-linear structures and patterns in time series data. These models can learn complex relationships within the data, enabling a more accurate capture of non-linearities present in the series, making them promising for inflation forecasting and analysis [10], [11].

Hybrid forecasting models have emerged to enhance traditional linear and non-linear techniques. These models amplify analytical capabilities and effectively mitigate prediction errors by combining a linear method with other non-linear approaches [12], [13].

The combination of these models allows for exploring their strengths: the ability of linear models to capture trends and linear patterns and the flexibility and capacity of non-linear models to capture non-linearities. This approach has shown promise as it provides better adaptation to the data, enhances result interpretation, increases robustness, and reduces the issue of overfitting. Combining linear and non-linear models in residual forecasting makes hybrid models a promising alternative in time series forecasting [14]–[16].

Residual modeling is one of the main challenges in creating an accurate hybrid system. Training machine learning models can be complex due to random fluctuations and noise in the residual series. These characteristics can bias the model and reduce its generalization capacity, leading to issues such as misspecification, underfitting, or overfitting [17], [18]. Additionally, more than a single nonlinear model may be required to capture time-varying fluctuations due to heteroscedasticity. In this context, using an ensemble of machine learning models shows promise, combining the predictions from multiple models to obtain a single final output [19].

The use of machine learning ensemble models emerges as an approach to tackle the challenges of residual modeling. This strategy combines predictions from multiple models to obtain a single final output [19], [20]. It has been widely applied in the literature and has shown a better ability to handle issues of overfitting and underfitting compared to individual models [21], [22]. Furthermore, decomposing the error into bias and variance reveals that the ensemble approach aims to reduce the variance of the error by combining different predictions [23].

This paper proposes a hybrid system that combines a linear statistical model with an ensemble of machine learning models for modeling residuals in the forecast of IPCA, the official inflation index in Brazil. The proposed method consists of three phases: (a) modeling the time series using a linear model, (b) modeling the residuals (original series - linear prediction) using an ensemble of non-linear models (bagging), and (c) combining the predictions from (a) and (b) through summation.

The remainder of this paper is structured as follows: the related works are presented in section II, the proposed method is explained in section III, the experiments are detailed in section IV and the conclusion is given in section VI.

II. RELATED WORK

The modeling of time series for forecasting can be represented in eq. 1, where z_t is the current time series and $z_{t-\tau}$ is the previously observed value, and f is the function that utilizes past values to predict future ones.

$$\hat{z}_t = f(z_{t-1}, z_{t-2}, \dots, z_{t-\tau}) \quad (1)$$

Although linear models, such as ARIMA, are widely used to estimate the inflation rate, they fail to capture nonlinear patterns in time series. For example, the ARIMA(p, d, q) model proposed by [8] is a combination of the AR(p) and MA(q) models with integration of order d .

The Akaike Information Criterion (AIC) evaluates the goodness of fit of a statistical model, considering the number of parameters and the likelihood of the data. The lower the AIC value, the better the fit and generalization capability for new data [24].

The ETS technique is based on the decomposition of a time series into three pillars - error, trend, and seasonality [25]. the ETS model has stood out for its effectiveness in forecasting

time series, such as inflation. Recent studies have focused on the ETS's ability to capture the volatility inherent in inflation. According to a study by Jere [26], the ETS method showed performance in predicting Zambia's inflation rate compared to other traditional approaches.

However, nonlinear forecasting models, such as MLP, SVR, and RF have been highlighted for their ability to capture complex relationships in nonlinear data [12], [27], [28].

The SVR model is an extension of the Support Vector Machine (SVM) model for regression. It utilizes the high-dimensional feature space and penalizes complexity through an additional term added to the error function. In the prediction of time series, the SVR model is represented in eq. 2:

$$f(x) = (w^T \cdot \phi(x)) + b \quad (2)$$

Here $f(x)$ is the prediction, w are the weights, b is the threshold, $\phi(\cdot)$ is a kernel function, and x is the time window. The minimization of the regularized risk function, as shown in eq. 3, is used to obtain the weight estimates w and threshold b :

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^l L(y_i, f(x)) \quad (3)$$

where $C > 0$ is the regularization factor controlling the trade-off between maximizing the margin and minimizing the training error.

The MLP is an artificial neural network model that utilizes multiple layers of neurons to learn complex relationships in data. Comprised of input layers, one or more hidden layers, and an output layer, each neuron calculates its output through a weighted combination of input values, followed by a non-linear activation function. MLP is widely applied in various machine learning domains because it can model non-linear relationships in data [27].

The output of an MLP with n hidden layers and m neurons in the output layer can be calculated as:

$$f(x) = \sigma(\mathbf{W}^{(n)} \cdot \sigma(\mathbf{W}^{(n-1)} \cdot \dots \sigma(\mathbf{W}^{(1)} \cdot \mathbf{x}))) \quad (4)$$

Here $\mathbf{x}$ represents the input vector, $\mathbf{W}_i$ are the weight matrices of the layers, and $\sigma(\cdot)$ is a non-linear activation function such as the sigmoid function or the ReLU function.

The Random Forest is a model based on decision trees that utilize an ensemble technique for prediction. It consists of a combination of multiple decision trees, where each tree is built from a random sample of the training data and a random subset of input variables. The final prediction is obtained by voting or averaging the individual tree predictions. Random Forest is known for its ability to handle high-dimensional data and its resistance to overfitting [28]. The prediction of the Random Forest shown in 5:

$$f(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^T f_t(\mathbf{x}) \quad (5)$$

Considering $f(\mathbf{x})$ the final prediction, T is the number of trees in the forest, and $f_t(\mathbf{x})$ is the prediction of the t -th tree.

In the general context, combining linear and non-linear functions in forecasting methodologies for time series has been shown to yield more accurate results compared to individual models [15], [29]–[31]. Hybrid models have been widely applied to macroeconomic data due to their improved performance [16], [32], [33]. The model combination can be achieved through ensemble approaches, where multiple models are trained individually, and their predictions are combined to generate the final forecast. Another approach to combining forecasts is through sequential hybrid systems, which incorporate linear forecasting, residual forecasting, and combination steps [34].

A practical approach to forecast combination requires the presence of diverse models. In the field of time series forecasting, it is common to employ Ensemble methods [31], [35] and residual hybrid systems to create combination approaches [12], [13], [16], [36].

As time series forecasting can be seen as a regression problem, it is common to aggregate the predictions of n models using either the mean or the median [37]. These methods are well-known for ensemble aggregation and can achieve similar results to more complex strategies [38]. Moreover, the mean and median aggregation strategies are non-trainable, requiring minimal computational effort. The following equation defines the mean aggregation:

$$\hat{Z}_{t+1} = \frac{1}{n} \sum_{i=1}^n (Z_t)^i \quad (6)$$

The proposed model forecasts the IPCA through a sequential hybrid system that utilizes the bagging approach in modeling the residuals of the linear model.

III. PROPOSED METHOD

The proposed model combines the ETS method with bagging for inflation forecasting in Brazil using the IPCA. This model consists of three main steps: linear forecasts, residual modeling, and a combination for the final forecast (ETS+Bagging). The goal is to leverage the advantages of both methods to improve the accuracy of IPCA predictions. The architecture of the proposed model is shown in Fig. 1.

A. Linear modeling forecasting

The first step involved performing the modeling of IPCA using the ETS method based on the assumption that there is a linear correlation between linear and nonlinear models [39]. The eq. 7 illustrates this relationship. Z_t represents the time series, L_t indicates the linear component, and N_t represents the nonlinear component.

$$Z_t = L_t + N_t \quad (7)$$

The best parameters for ETS were selected using the Akaike Information Criterion (AIC) [25]. The equation calculates the AIC:

$$\text{AIC} = -2 \log(L) + 2k, \quad (8)$$

Where L is the likelihood function of the model and k is the number of estimated parameters. Each fitted model is assigned the resulting AIC value, and the model with the lowest AIC value is selected as the best model. This approach penalizes more complex models, avoiding overfitting the data [40].

Once the linear predictions L_t are made, the residual (or error) series E_t is computed as per eq. 9:

$$E_t = Z_t - \hat{L}_t \quad (9)$$

B. Residual modeling forecasting

In the second step, we utilized the bagging technique with three different models: SVR, MLP, and RF, to perform nonlinear modeling of the residuals [41], [42]. Each model was individually trained, optimizing their hyperparameters through a grid search technique [13]. This approach involves testing various combinations of predefined hyperparameters and evaluating their performance based on an evaluation metric, such as the Mean Squared Error (MSE).

After hyperparameter optimization and model training, we applied the bagging technique by creating multiple instances of each base model (SVR, MLP, and RF). Each instance was trained with a random sample from the training dataset. Subsequently, the predictions from each model were combined using the arithmetic mean, resulting in a more stable and accurate final prediction. This nonlinear final prediction is represented as $\hat{N}_t$ [43].

C. Final combination of forecasts

The last step performs the final combination of forecasts obtained from the linear + Bagging approach. The linear model used was ARIMA, while the nonlinear forecasts were generated by the outputs of the instances of the models (SVR, MLP, and RF). The final forecast (ARIMA+Bagging) is shown in eq. 10.

$$\hat{Z}_t = \hat{L}_t + \hat{N}_t \quad (10)$$

IV. EXPERIMENTAL SETUP

The IPCA was used to analyze inflation. The Inflation data for IPCA in Brazil was extracted from IBGE [44], covering the period from February 1999 to March 2023, totaling 290 records. The database was divided into training (60%), validation (20%), and testing (20%) sets.

The ARIMA model was generated using the "pmdarima" package in Python, with parameter optimization based on the AIC criterion [45].

For the ETS model 'statsmodels' library, in Python, specifically the 'ExponentialSmoothing' class. First, we define the model's components, such as error, trend, and seasonality, and fit the model to the observed data. Then, the model parameters are estimated based on the data properties during the fitting process [46].

Non-linear models such as SVR, MLP, and RF were used to predict the residuals. These models were selected based

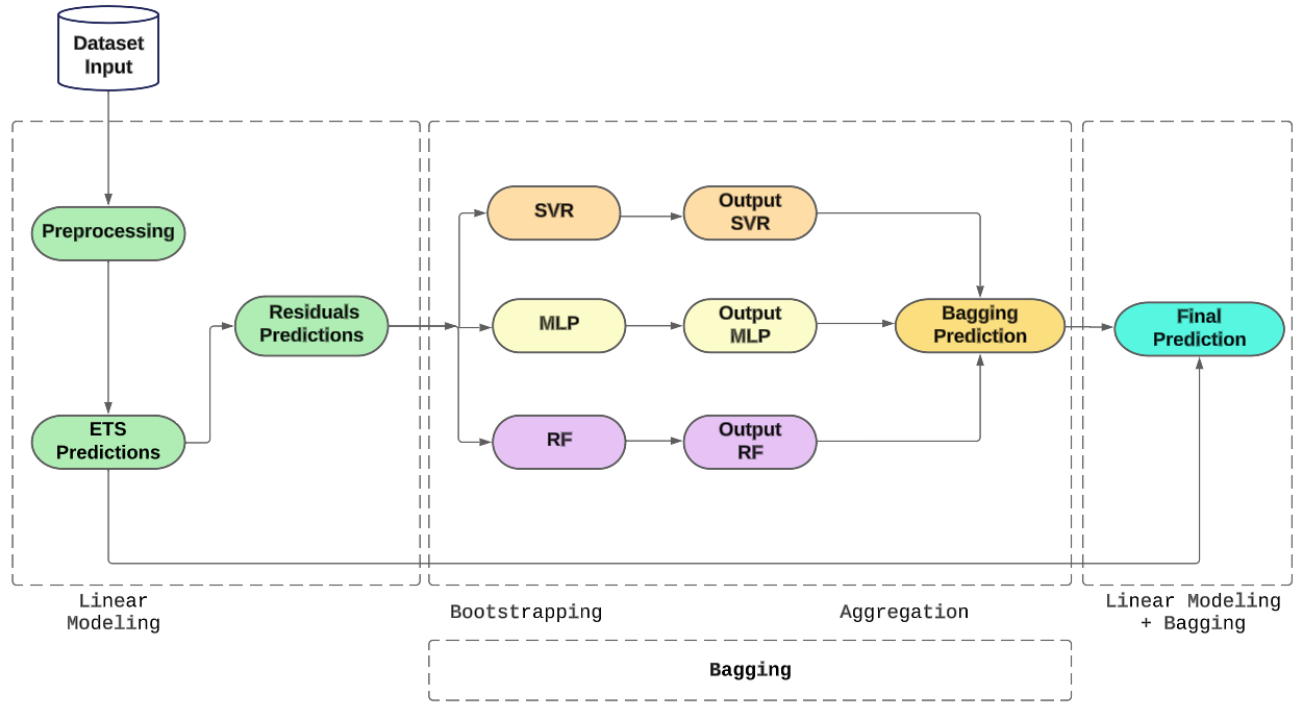


Fig. 1. Proposed hybrid system architecture

on previous research that applies hybrid systems in inflation forecasting [16], [47]–[50].

The best hyperparameters for the SVR, MLP, and RF regressors were searched using a grid search with the GridSearchCV library from scikit-learn [30], [37]. Then, different combinations of hyperparameters were tested for each regressor, as detailed in Table I.

TABLE I
PARAMETER VALUES USED IN THE SEARCH PROCESS FOR ALL METHODS

Model	Parameters	Option
ARIMA	p, d, q	Hyndman's Method [46]
ETS	<i>additive, multiplicative</i>	Hyndman's Method [46]
MLP	Input (Lag)	2, 5, 10, 15, 20
	Algorithm	Backpropagation
	Activation Function	Sigmoid
	Number of Hidden Layer Nodes	2, 5, 10, 15, 20
SVR	Input (Lag)	2, 5, 10, 15, 20
	Kernel	Linear, Poly, RBF, and Sigmoid
	γ	1, 0.1, 0.01, 0.001
	C	0.1, 1, 100, 1000, 10000
	ϵ	0.1, 0.01, 0.001
RF	Number of estimators	50, 100, 200, 500
	criterion	MSE
	Max depth	3, 5, 7, 9

Finally, the best configuration was selected based on the mean squared error (MSE) on the validation set. The final predictions of each method were evaluated using the metrics MSE, MAPE, MAE, and RMSE, as presented in Table II.

The MSE penalizes more significant prediction errors more intensely. At the same time, the RMSE has the same interpretation as the MSE but on the original scale of the data, facilitating the understanding of the average error size. The MAE provides a direct measure of the average error

size, regardless of the direction, while the MAPE provides a percentage measure of the error relative to the time series. It is important to note that the MAPE can be affected by values close to zero, leading to inconsistent results or distortions in accuracy assessment [24].

V. RESULTS AND DISCUSSION

To evaluate and compare different approaches, we conducted tests using thirteen relevant methods mentioned in the literature. These tests were performed on the same dataset used in this study. In addition, the proposed hybrid model, named ETS+Bagging, as compared to other widely used models in the literature, such as ARIMA, ETS, MLP, SVR, RF, ARIMA+SVR, ARIMA+MLP, ARIMA+RF, ARIMA+Bagging, ETS+SVR, ETS+MLP, and ETS+RF.

The results for current forecasts as presented in Table II. Analyzing the MSE, RMSE, and MAE, we found that the ETS model had the lowest performance among all tested models due to its limitation in mapping nonlinear patterns in the time series data. On the other hand, the SVR, MLP, and RF models, which can capture nonlinear mappings, outperformed the ARIMA and ETS models.

However, when considering the MAPE metric, the RF model had the worst performance among all tested models. Surprisingly, the ETS model outperformed the nonlinear models in the MAPE metric, indicating a lower percentage error relative to the actual values. It is essential to highlight that combining the linear model with the Bagging technique showed an improvement in the performance of the ARIMA

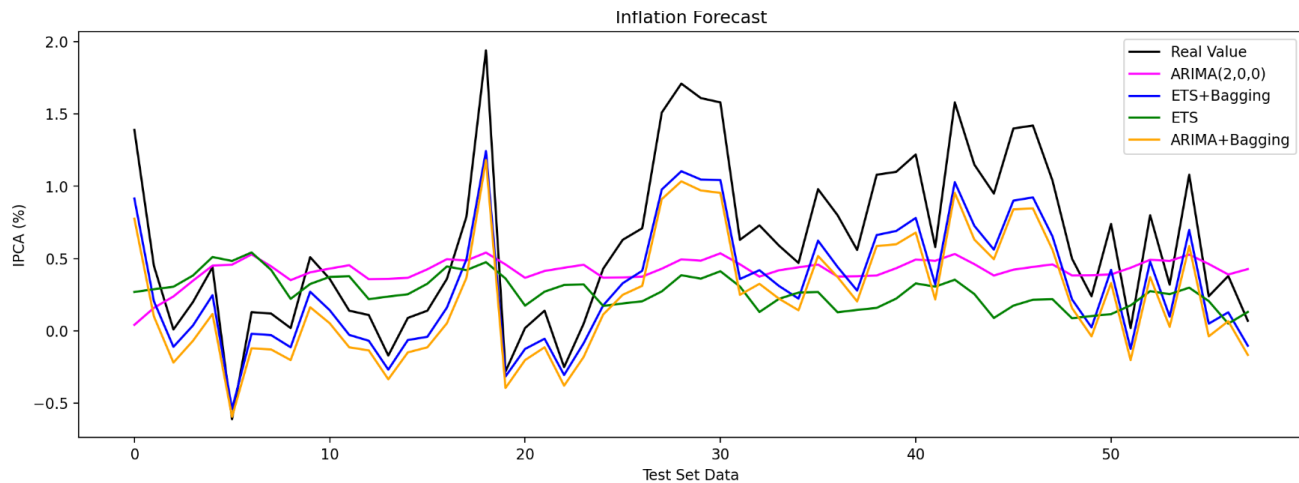


Fig. 2. Forecasts of the proposed method on the test set .

TABLE II
PERFORMANCE OF THE PROPOSED METHOD COMPARED TO INDIVIDUAL
MODELS AND HYBRID SYSTEMS

Models	MSE	RMSE	MAE	MAPE
ARIMA	0.3466	0.5887	0.4654	2.9707
ETS	0.4418	0.6647	0.5280	1.8851
SVR	0.2386	0.4885	0.3609	2.2783
MLP	0.2410	0.4909	0.3599	2.3041
RF	0.3295	0.5740	0.4504	3.9057
ARIMA + Bagging	0.1608	0.4010	0.3709	1.8163
ARIMA + SVR	0.1656	0.4070	0.3103	0.4814
ARIMA + MLP	0.1723	0.4151	0.3246	0.5507
ARIMA + RF	0.1775	0.4213	0.3304	0.5387
ETS + Bagging	0.1044	0.3230	0.2874	1.1806
ETS + SVR	0.1763	0.4199	0.3336	0.6295
ETS + MLP	0.1740	0.4172	0.3265	0.6163
ETS + RF	0.1778	0.4217	0.3315	0.5639

and ETS models. As seen in the comparative analysis of the methods in Fig. 2.

Furthermore, the proposed hybrid model (ETS+Bagging) better fits the data used in this study, with lower errors according to the MSE, RMSE, and MAE metrics. However, the ARIMA+SVR hybrid model stood out in the MAPE metric.

VI. CONCLUSIONS

This study presents a hybrid system with bagging to improve the forecast of the official inflation index in Brazil, the IPCA. The proposal combines a linear approach (ETS) with the bagging technique applied to residual forecasting. Three nonlinear models are used in the bagging process: SVR, MLP, and RF. Then, these models are combined through an arithmetic mean, resulting in a final forecast incorporating both the linear component and the bagging approach. The analysis period covers from February 1999 to March 2023, totaling 290 monthly frequency records. The experimental results reveal that the proposed method (ETS+Bagging) outperformed individual and hybrid models. In addition, the bagging approach improved the results compared to the ARIMA and ETS models in all error metrics, indicating its effectiveness

in capturing nonlinear patterns in the residuals and allowing better adaptation to the diverse characteristics and patterns of inflation data.

For future work, consider feature selection to improve the model's performance further. Additionally, exploring other nonlinear methods, such as XGBoost, and experimenting with different combinations of weights in the bagging process could enhance the robustness of the proposed method. These areas hold promise for future investigations to enhance IPCA forecasts' quality further.

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